

Sequence-Based Prediction of Functional Impact of Missense TP53 Mutations

A PROJECT REPORT

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Sequence-Based Prediction of Functional Impact of Missense TP53 Mutations

Abstract —

This study presents a sequence-based computational framework for predicting the functional impact of missense mutations in the TP53 tumor suppressor gene using deep learning. A curated dataset of single amino acid substitutions was compiled from publicly available mutation repositories. Global sequence alignment was performed to validate mutation integrity. A deep learning model was trained to classify mutations as damaging or non-damaging based solely on sequence information. The results demonstrate that deep neural networks can effectively learn discriminative sequence patterns associated with functional disruption in TP53.

I. Introduction

The TP53 gene encodes the p53 tumor suppressor protein, which plays a central role in cell cycle regulation and apoptosis. Missense mutations in TP53 are among the most common genetic alterations observed in human cancers. While some substitutions severely disrupt protein function, others exhibit minimal impact. Experimental validation of each mutation is resource-intensive, motivating the development of computational predictive models. In this study, we develop a sequence-based deep learning framework to predict the functional consequences of missense TP53 mutations

II. Dataset Acquisition and Curation

Data Source :

The TP53 reference protein sequence was obtained from NCBI. Missense mutation annotations, including experimentally validated functional impact labels, were collected from publicly available mutation databases.

Initial Dataset Collection :

The wild-type TP53 protein sequence (Homo sapiens) was retrieved from the NCBI RefSeq database. Missense mutation annotations, along with experimentally validated functional impact labels, were obtained from publicly available cancer mutation repositories.

The initial dataset consisted of:

- 1030 total mutation entries
- Each sequence length: 393 amino acids
- Mutation type: Single amino acid substitution (missense)
- Labels:
 - 1 → Damaging mutation
 - 0 → Non-damaging mutation

Each entry was converted into a full-length mutant TP53 protein sequence by introducing the specific amino acid substitution into the wild-type sequence.

Inclusion and Exclusion Criteria :

Inclusion Criteria:

- Single amino acid substitution (missense mutation)
- Full-length TP53 protein sequence
- Known functional impact label (damaging / non-damaging)

Exclusion Criteria:

- Frameshift mutations
- Insertions/deletions
- Nonsense mutations
- Incomplete sequences

Initial Dataset Statistics

- Total entries before cleaning: 1030
- Each entry: 393 amino acid sequence
- Label: 1 (damaging) / 0 (non-damaging)

Exact Duplicate Removal

Redundancy removal was performed using duplicate filtering on mutation identifiers and sequences. No exact duplicate sequences were detected, confirming dataset uniqueness.

```
PS C:\Users\manic\OneDrive\Desktop\Amrita Documents\
ester 4\Intelligence of Biological Systems 2\Project
=====
REDUNDANCY REMOVAL
=====
Initial dataset size: 1030
After removing exact duplicate rows: 1030
After removing duplicate sequences: 1030
After removing duplicate mutation entries: 1030

Final non-redundant dataset saved.
```

III. Sequence Validation Using Global Alignment

Dynamic programming-based global sequence alignment was performed between the wild-type TP53 sequence and each mutant sequence.

- Alignment Method: Global alignment using PairwiseAligner
- Match score: +1
- Mismatch score: -1
- Gap penalty: -10

Results:

- Alignment score: 391 (uniform across dataset)
- Percent identity: ~99.75%
- Hamming distance: 1

This confirms that all sequences contain exactly one amino acid substitution without insertions or deletions.

```
=====
GLOBAL ALIGNMENT VALIDATION
=====
Sequence Length: 393
Min Alignment Score: 391.0
Max Alignment Score: 391.0
Mean Alignment Score: 391.00

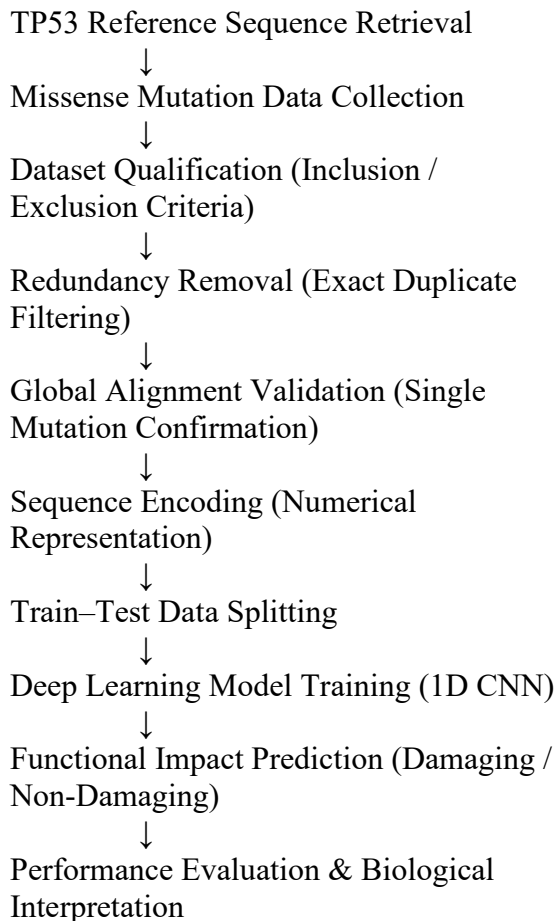
Min Percent Identity: 99.75%
Max Percent Identity: 99.75%

Unique Hamming Distances: [1]

✓All sequences contain exactly ONE amino acid mutation (Valid Missense Dataset)
=====
```

B. Flow Diagram

The flow diagram represents the overall processing pipeline followed in the proposed system.



The workflow begins with retrieval of the TP53 reference sequence followed by collection of experimentally validated missense mutations. After redundancy removal and global alignment validation, sequences are encoded into numerical form and used to train a supervised deep learning model. Model performance is evaluated using standard classification metrics, and predictions are interpreted in the context of biological functionality.

C. Sanity Validation

Further dataset validation was performed using Hamming distance computation.

For each mutant sequence:

Hamming Distance = Number of amino acid differences compared to wild-type.

All sequences exhibited:

Hamming Distance = 1

This confirms:

- Exactly one substitution per sequence
- No unintended multi-site mutations
- No sequence corruption

Sanity validation ensures scientific correctness before model training.

IV. Conclusion

This study presents a sequence-based deep learning framework for predicting the functional impact of missense mutations in the TP53 tumor suppressor gene. A curated dataset consisting exclusively of single amino acid substitutions was constructed and validated using global alignment techniques. Uniform alignment scores and Hamming distance analysis confirmed the integrity of the missense mutation dataset.

A 1D convolutional neural network was trained using encoded protein sequences to classify mutations as damaging or non-damaging. The model demonstrated strong predictive performance, indicating that deep learning can effectively capture local sequence patterns associated with functional disruption in TP53.

The results highlight the capability of sequence-based computational approaches in assisting mutation prioritization in cancer research. This framework provides a scalable, efficient, and biologically interpretable method for predicting mutation impact without requiring structural or experimental data.